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Passivation structure for a thin film transistor

Abstract

In some embodiments, the present disclosure relates to a device that includes an active layer, a gate electrode, a passivation structure, a source contact, and a drain contact arranged over a substrate. The gate electrode is arranged over the substrate and is spaced apart from the active layer by a gate dielectric layer. The passivation structure is arranged over the active layer. The source contact extends through the passivation structure and contacts the active layer. The drain contact extends through the passivation structure and contacts the active layer. The passivation structure is hydrophobic.

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Background/Summary

BACKGROUND
(1) As technology advances at a rapid pace, engineers work to make devices smaller, yet more complex to improve and develop electronic devices that are more efficient, more reliable, and have more capabilities. One way to achieve these goals is by improving the design of transistors, as electronic devices comprise a plethora of transistors that together, carry out the function of the device. Overall electronic device performance may benefit from transistors that, for example, are smaller, consume less power, and have faster switching speeds.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various

features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. **1**A illustrates a cross-sectional view of some embodiments of a bottom-gate thin film transistor comprising a passivation structure arranged over an active layer and comprising silicon carbide.

(3) FIG. **1**B illustrates a cross-sectional view of some embodiments of the bottom-gate thin film transistor of FIG. **1**A in a back-end-of-line interconnect structure.

(4) FIG. **2** illustrates a cross-sectional view of some other embodiments of a bottom-gate thin film transistor comprising a passivation structure arranged over an active layer, comprising two layers, and comprising silicon carbide.

(5) FIG. **3** illustrates a cross-sectional view of some embodiments of a top-gate thin film transistor comprising a passivation structure laterally between a source contact, a drain contact, and a gate electrode structure, wherein the passivation structure comprises silicon carbide.

(6) FIG. **4**A illustrates a cross-sectional view of some other embodiments of a top-gate thin film transistor comprising a passivation structure laterally between a source contact, a drain contact, and a gate electrode structure, wherein the passivation structure comprises two layers and comprises silicon carbide.

(7) FIG. **4**B illustrates a cross-sectional view of some embodiments of the top-gate thin film transistor of FIG. **4**A in a back-end-of-line interconnect structure.

(8) FIGS. **5-11**B illustrate cross-sectional views of some embodiments of a method of forming a bottom-gate thin film transistor having a passivation structure that mitigates water and oxygen from the surrounding environment from damaging an active layer arranged beneath the passivation structure.

(9) FIG. **12** illustrates a flow diagram of some embodiments corresponding to the method of FIGS. **5-11**B.

(10) FIGS. **13-19**B illustrate cross-sectional views of some embodiments of a method of forming a top-gate thin film transistor having a passivation structure that mitigates water and oxygen from the surrounding environment from damaging an active layer arranged beneath the passivation structure.

(11) FIG. **20** illustrates a flow diagram of some embodiments corresponding to the method of FIGS. **13-19**B.

DETAILED DESCRIPTION

(12) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(13) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(14) In some embodiments, a transistor, for example, thin film transistor (TFT) comprises an active layer arranged over a substrate. The active layer may be turned “ON” such that mobile charge

carriers flow through the active layer when signals (e.g., voltage, current) are applied to source, drain, and gate contacts of the TFT. The substrate of a TFT is often not a conductive material, but instead is a supportive substrate for the active layer of the TFT. For example, in some instances, the substrate is a transparent material such as glass because TFTs may be used in optical applications such as for liquid-crystal displays. The active layer is thinner than a typical semiconductor substrate used in other transistors, such as planar metal oxide semiconductors field effect transistors (MOSFET). In some instances, the active layer comprises a semiconductor material that is transparent such as, for example, indium gallium zinc oxide (IGZO), amorphous silicon, or some other suitable material.

(15) In some embodiments of a top-gate TFT, the gate electrode is arranged over the active layer and directly between a source contact and a drain contact. The gate electrode, the source contact, and the drain contact may be spaced apart from one another by a passivation layer. In some embodiments of a bottom-gate TFT, the gate electrode is arranged below the active layer and the source contact and the drain contact are arranged over the active layer. The source contact and the drain contact are spaced apart from one another by a passivation structure. The passivation structure of a TFT covers the active layer and comprises a material used to mitigate or prevent water and oxygen from the environment from diffusing into the active layer and degrading the TFT. However, oftentimes, the passivation structure comprises silicon dioxide which requires a low processing temperature (e.g., less than 300 degrees Celsius) to prevent damage to the silicon dioxide. Thus, if the passivation structure becomes damaged when exposed to high temperatures (e.g., greater than 300 degrees Celsius) during processing, the passivation layer may fail in effectively protecting the active layer, thereby reducing the TFT reliability. Further, silicon dioxide may be hydrophilic and have a low film density which would allow moisture to enter the passivation structure and diffuse into the active layer.

(16) Various embodiments of the present disclosure relate to forming a top-gate or a bottom-gate TFT with a passivation structure arranged over the active layer that has one or more layers comprising materials that are more reliable in mitigating the diffusion of water and oxygen into the active layer than silicon dioxide. In some embodiments, at least an upper portion of the passivation structure comprises oxygen-doped silicon carbide (ODC), nitrogen-doped silicon carbide (NDC), or a mixture thereof. In such embodiments, ODC and NDC are more effective in preventing moisture (e.g., water) and oxygen from the environment from diffusing into the active layer than silicon dioxide because ODC and NDC are hydrophobic and have a higher film density than silicon dioxide. Further, ODC and NDC are commonly used in back-end-of-line metal processing, and thus, incorporating ODC and NDC into the passivation structure of the TFT may be easier because equipment and precursors used for ODC and NDC processing is already available. By incorporating ODC and/or NDC materials into the passivation structure of a TFT, damage to the TFT from the environment is mitigated thereby increasing the longevity and reliability of the TFT.

(17) FIG. **1**A illustrates a cross-sectional view **100**A of some embodiments of an integrated chip comprising a bottom-gate thin film transistor (TFT).

(18) The bottom-gate TFT of FIG. **1**A includes a gate electrode **104** arranged over a substrate **102**. Further, in some embodiments, the gate electrode **104** is arranged below an active layer **108**. In some embodiments, the gate electrode **104** is spaced apart from the active layer **108** by a gate dielectric layer **106**. Because the gate electrode **104** is arranged between the active layer **108** and the substrate **102**, the TFT of FIG. **1**A is called a “bottom-gate” TFT. In some embodiments, the substrate **102** is a supportive, non-semiconducting substrate. For example, in some embodiments, the substrate **102** comprises a glass, or some other rigid, supportive material.

(19) In some embodiments, the active layer **108** comprises a semiconductor material. In some embodiments, the substrate **102** and the active layer **108** comprise transparent materials because the bottom-gate TFT may be used in some optical applications. Examples of transparent materials for the active layer **108** include indium gallium zinc oxide, indium tin oxide, indium gallium oxide,

indium zinc oxide, indium gallium zinc tin oxide, indium oxide, or some other suitable material. Further, examples of transparent materials for the gate dielectric layer **106** include hafnium oxide, zirconium oxide, aluminum oxide, lanthanum oxide, silicon dioxide, silicon nitride, a combination thereof, or some other suitable dielectric material. Examples of transparent materials for the gate electrode **104** include indium tin oxide, doped zinc oxide, or some other suitable transparent, conductive material. It will be appreciated that in some other embodiments, the bottom-gate TFT is not used in optical applications and thus, does not comprise as many transparent materials.

(20) In some embodiments, a passivation structure **114** is arranged over the active layer **108**, and a source contact **110** and a drain contact **112** extend through the passivation structure **114** to contact the active layer **108**. Thus, in some embodiments, the bottom-gate TFT is turned “ON” when signals (e.g., voltage, current) are applied to the gate electrode **104**, the source contact **110**, and the drain contact **112** that exceed a threshold voltage of the bottom-gate TFT such that mobile charge carriers (e.g., holes or electrons) flow through the active layer **108**. In some embodiments, the passivation structure **114** protects the active layer **108** from damage from the environment. For example, the passivation structure **114** mitigates moisture (e.g., water) and oxygen from the surrounding environment of the bottom-gate TFT from diffusing into the active layer **108** which may damage the active layer **108** and thus, reduce the reliability and/or longevity of the bottom-gate TFT.

(21) In some embodiments, an effective material for the passivation structure **114** has a high film density and is hydrophobic to prevent moisture from entering into the passivation structure **114** and into the active layer **108**. Further, the passivation structure **114** comprises an insulator and/or dielectric material to electrically isolate the source contact **110** from the drain contact **112**. In some embodiments, the passivation structure **114** has a higher film density than that of silicon dioxide, for example. In some embodiments, the film density of the passivation structure **114** is measured using X-ray reflective (XRR) measurements. An XRR measurement measures a relationship between properties of incident X-rays and reflected X-rays to determine the film density.

(22) In some embodiments, the passivation structure **114** comprises, for example, oxygen-doped silicon carbide, nitrogen-doped silicon carbide, a mixture thereof, or some other suitable material that has a high film density, is hydrophobic, and is electrically insulating. Further, in some embodiments, the passivation structure **114** comprises a material that can maintain these aforementioned properties when exposed to high temperatures (e.g., greater than 250 degrees Celsius) during processing. In some embodiments, the passivation structure **114** has a thickness in a range of between, for example, approximately 5 nanometers and approximately 1000 nanometers. In some embodiments, the passivation structure **114** comprises a first passivation layer **116** that continuously extends in the vertical direction from the active layer **108** to a top of the source and drain contacts **110**, **112**. Thus, the passivation structure **114** comprises one or more materials that mitigate moisture (e.g., water), oxygen, and/or other vapors/gases in the environment from reaching the active layer **108** to increase the reliability of the bottom-gate TFT.

(23) FIG. **1**B illustrates a cross-sectional view **100**B of some embodiments of a back-end-of-line (BEOL) interconnect structure comprising the bottom-gate TFT of FIG. **1**A.

(24) In some embodiments, a first bottom-gate TFT **101***a* may be arranged within a back-end-of-line (BEOL) interconnect structure **120** over the substrate **102**. Thus, in some embodiments, the gate electrode **104** may not be arranged directly on the substrate **102**. In some embodiments, the back-end-of-line (BEOL) interconnect structure **120** may comprise, for example, interconnect vias **122** and interconnect wires **124** embedded within interconnect dielectric layers **128** and etch stop layers **126**. In some embodiments, the BEOL interconnect structure **120** further comprises a bond pad **130** arranged over a topmost interconnect wire **124** or interconnect via **122** and comprises a solder bump **132** arranged over the bond pad **130**.

(25) In some embodiments, the interconnect vias **122**, the interconnect wires **124**, the bond pads **130**, and the solder bumps **132** comprise conductive materials such as, for example, copper,

aluminum, tungsten, tantalum, titanium, tantalum nitride, titanium nitride, or the like. In some embodiments, the interconnect dielectric layers **128** and the etch stop layers **128** may comprise a dielectric material, such as, for example, a nitride (e.g., silicon nitride, silicon oxynitride), a carbide (e.g., silicon carbide), an oxide (e.g., silicon oxide), borosilicate glass (BSG), phosphoric silicate glass (PSG), borophosphosilicate glass (BPSG), a low-k oxide (e.g., a carbon doped oxide, SiCOH), or the like.

(26) In some embodiments, the first bottom-gate TFT **101***a* may be arranged within a lower portion of the BEOL interconnect structure **120**, and interconnect vias **122** may be electrically coupled to the source contact **110**, the drain contact **112**, and the gate electrode **104** of the first bottom-gate TFT **101***a*. In some embodiments, a second bottom-gate TFT **101***b* may be arranged in a middle portion of the BEOL interconnect structure **120**. Further, in some embodiments, the second bottom-gate TFT **101***b* may have a gate electrode **104** arranged directly over a substrate **102** of the second bottom-gate TFT **101***b* or some other supportive structure. In some embodiments, a third bottom-gate TFT **101***c* may be arranged within an upper portion of the BEOL interconnect structure **120**. In some embodiments, the gate electrode **104** may be arranged on the etch stop layer **126**, and the gate electrode **104** may be coupled to the interconnect wire **124** below the etch stop layer **126**. In some other embodiments, the gate electrode **104** may be arranged directly on the interconnect wire **124**. In some such embodiments, the source contact **110** and the drain contact **112** of the third bottom-gate TFT **101***c* are coupled to the solder bumps **132** through the bond pads **130**, interconnect wires **124**, and interconnect vias **122**.

(27) In some embodiments, a fourth bottom-gate TFT **101***d* may comprise the active layer **108**, the first passivation layer **116**, the source contact **110**, and the drain contact **112**. In some such embodiments, the interconnect wire **124** may serve as the gate electrode of the fourth bottom-gate TFT **101***d*, and the etch stop layer **126** may serve as the gate dielectric layer of the fourth bottom-gate TFT **101***d*.

(28) Thus, the cross-sectional view **100**B of FIG. **1**B illustrates various areas that a bottom-gate TFT may be arranged in a BEOL interconnect structure **120**. In some embodiments, high temperatures are used during the formation of the BEOL interconnect structure **120**. Thus, because of the bottom-gate TFTs (e.g., **101***a-d*) comprise a passivation structure (**114** of FIG. **1**A) that comprise a material that can maintain its properties when exposed to high temperatures (e.g., greater than 250 degrees Celsius), the bottom-gate TFTs (e.g., **101***a-d*) may be integrated into the BEOL interconnect structure **120** without suffering damage. Further, because of the small size of the bottom-gate TFTs (e.g., **101***a-d*), the bottom-gate TFTs (e.g., **101***a-d*) may be integrated in the BEOL interconnect structure **120** without increasing the height of the BEOL interconnect structure **120**.

(29) FIG. **2** illustrates a cross-sectional view **200** of some other embodiments of a bottom-gate TFT, wherein the passivation structure comprises more than one layer.

(30) In some other embodiments, the passivation structure **114** comprises a second passivation layer **202** arranged between the first passivation layer **116** and the active layer **108**. In some such embodiments, the first and second passivation layers **116**, **202** laterally surround the source contact **110** and the drain contact **112**. In some embodiments, at least the first passivation layer **116**, which is an upper portion of the passivation structure **114**, comprises a material that has a high film density, is hydrophobic, and is electrically insulating, such as, for example, oxygen-doped silicon carbide, nitrogen-doped silicon carbide, or the like. In some embodiments, the second passivation layer **202** may also comprise a material that has a high film density, is hydrophobic, and is electrically insulating, such as oxygen-doped silicon carbide, nitrogen-doped silicon carbide, or the like.

(31) In other embodiments, the second passivation layer **202** may comprise a material that has a lower film density and/or is less hydrophobic than the first passivation layer **116**. For example, in such other embodiments, the second passivation layer **202** may comprise, for example, silicon

dioxide, silicon nitride, aluminum oxide, hafnium oxide, hafnium zirconium oxide, or the like. Thus, in some embodiments, the first passivation layer **116** may comprise oxygen-doped silicon carbide and/or nitrogen-doped silicon carbide to mitigate moisture (e.g., water) and oxygen from diffusing into the second passivation layer **202** and into the active layer **108** to increase the reliability of the bottom-gate TFT.

(32) Further, in some embodiments, the source contact **110** may have a bottom surface **110***b* that is arranged below a topmost surface **108***t* of the active layer **108**. Similarly, in some embodiments, the drain contact **112** may have a bottom surface **112***b* that is arranged below the topmost surface **108***t* of the active layer **108**. In some such embodiments, the source and drain contacts **110**, **112** are formed by first forming openings in the first and second passivation layers **116**, **202**. During the formation of the openings, portions of the active layer **108** may be removed due to, for example, over etching, in some embodiments, Thus, in some such embodiments, the source and drain contacts **110**, **122**, which are formed within the openings, have bottommost surfaces, **110***b*, **112***b*, respectively, that are arranged below the topmost surface **108***t* of the active layer **108**. In some other embodiments, the active layer **108** is substantially resistant to removal by the removal process used to form the openings, and thus, the bottommost surface **110***b* of the source contact **110** and the bottommost surface **112***b* of the drain contact **112** directly contact the topmost surface **108***t* of the active layer **108**.

(33) FIG. **3** illustrates a cross-sectional view **300** of some embodiments of a top-gate TFT.

(34) The integrated chip of FIG. **3** includes a gate electrode **104** arranged over the active layer **108**, such that the TFT of FIG. **3** is called a “top-gate” TFT. In some such embodiments, the gate dielectric layer **106** is arranged between the gate electrode **104** and the active layer **108**. Further, in some embodiments, the gate electrode **104** and the gate dielectric layer **106** extend through the passivation structure **114**. In some embodiments, the gate electrode **104** and the gate dielectric layer **106** are arranged laterally between the source contact **110** and the drain contact **112**. Thus, in some embodiments, the gate electrode **104** and the gate dielectric layer **106** are separated from the source contact **110** and the drain contact **112** by the passivation structure **114**. In some embodiments, the passivation structure **114** comprises the first passivation layer **116**.

(35) FIG. **4**A illustrates a cross-sectional view **400**A of some embodiments of a top-gate TFT, wherein the passivation structure comprises more than one layer.

(36) In some other embodiments of a top-gate TFT, the passivation structure **114** comprises the second passivation layer **202** arranged between the first passivation layer **116** and the active layer **108**. In some embodiments, at least the first passivation layer **116**, which makes up the upper portion of the passivation structure **114** comprises a material that has a high film density, is hydrophobic, and is electrically insulating to prevent moisture (e.g., water) and oxygen from the environment from damaging the active layer **108**.

(37) FIG. **4**B illustrates a cross-sectional view **400**B of some embodiments of a back-end-of-line (BEOL) interconnect structure comprising the top-gate TFT of FIG. **4**A.

(38) In some embodiments, a first top-gate TFT **401***a* may be arranged within a lower portion of the BEOL interconnect structure **120**, and interconnect vias **122** may be electrically coupled to the source contact **110** and the drain contact **112**. In some embodiments, a second top-gate TFT **401***b* may be arranged in a middle portion of the BEOL interconnect structure **120**. In some embodiments, the second top-gate TFT **401***b* may have an active layer **108** arranged over a substrate **102** of the second top-gate TFT **401***b* or arranged over some other supportive structure. In some embodiments, an interconnect via **122** is also coupled to the gate electrode **104** of the second top-gate TFT **401***b*. In some embodiments, a third top-gate TFT **401***c* is arranged in an upper portion of the BEOL structure **120**, and the active layer **108** of the third top-gate TFT **401***c* may be arranged directly on the etch stop layer **126**.

(39) Thus, the cross-sectional view **400**B of FIG. **4**B illustrates various areas that a top-gate TFT may be arranged in a BEOL interconnect structure **120**. In some embodiments, high temperatures

are used during the formation of the BEOL interconnect structure **120**. Thus, because of the top-gate TFTs (e.g., **401***a*-*c*) comprise a passivation structure (**114** of FIG. **4**A) that comprise a material that can maintain its properties when exposed to high temperatures (e.g., greater than 250 degrees Celsius), the top-gate TFTs (e.g., **401***a*-*c*) may be integrated into the BEOL interconnect structure **120** without suffering damage. Further, because of the small size of the top-gate TFTs (e.g., **401***a*-*c*), the top-gate TFTs (e.g., **401***a*-*c*) may be integrated in the BEOL interconnect structure **120** without increasing the height of the BEOL interconnect structure **120**.

(40) It will be appreciated that in some other embodiments, a combination of bottom-gate TFTs (e.g., **101***a*-*d* of FIG. **1**B) and top-gate TFTs (e.g., **401***a*-*c*) may be integrated into a same BEOL interconnect structure **120**.

(41) FIGS. **5**-**11**B illustrate cross-sectional views **500**-**1100**B of some embodiments of a method of forming a bottom-gate thin film transistor (TFT) having a passivation structure comprising silicon carbide. Although FIGS. **5**-**11**B are described in relation to a method, it will be appreciated that the structures disclosed in FIGS. **5**-**11**B are not limited to such a method, but instead may stand alone as structures independent of the method.

(42) As shown in cross-sectional view **500** of FIG. **5**, a substrate **102** is provided. In some embodiments, the substrate **102** comprises a supportive material, such as, for example, glass. In some embodiments, the substrate **102** comprises a transparent material suitable for optical applications, whereas in other embodiments, the substrate **102** comprises a translucent or opaque material. Unlike many other types of transistors, in some embodiments, the substrate **102** is not a semiconductor material and is thus, used for support instead of also for electrical reasons. In some embodiments, a gate electrode **104** is formed over the substrate **102**. In some embodiments, the gate electrode **104** may be formed by way of a deposition process (e.g., physical vapor deposition (PVD), chemical vapor deposition (CVD), plasma-enhanced CVD (PE-CVD), atomic layer deposition (ALD), sputtering, etc.). In some embodiments, the gate electrode **104** comprises a conductive material. In some embodiments, the gate electrode **104** also comprises a transparent material for use in optical applications. For example, in some embodiments, the gate electrode **104** may comprise indium tin oxide, doped zinc oxide, or some other suitable transparent, conducting material. In some embodiments, the gate electrode **104** is formed to have a thickness in a range of between, for example, approximately 3 nanometers and approximately 90 nanometers.

(43) As shown in cross-sectional view **600** of FIG. **6**, a gate dielectric layer **106** is formed over the gate electrode **104**. In some embodiments, the gate dielectric layer **106** is formed by way of a deposition process (e.g., PVD, CVD, PE-CVD, ALD, sputtering, spin-on, etc.). In some embodiments, the gate dielectric layer **106** comprises a dielectric material. In some embodiments, the gate dielectric layer **106** also comprises a transparent material for optical applications. For example, in some embodiments, the gate dielectric layer **106** comprises hafnium oxide, zirconium oxide, aluminum oxide, lanthanum oxide, silicon dioxide, silicon nitride, a combination thereof, or some other suitable dielectric material. In some embodiments, the gate dielectric layer **106** is formed to have a thickness in a range of between, for example, approximately 3 nanometers and approximately 90 nanometers.

(44) As shown in cross-sectional view **700** of FIG. **7**, an active layer **108** is formed over the gate dielectric layer **106**. In some embodiments, the active layer **108** is formed by way of a deposition process (e.g., PVD, CVD, PE-CVD, ALD, sputtering, spin-on, etc.). In some embodiments, the active layer **108** comprises a semiconductor material. In some embodiments, the active layer **108** also comprises a transparent material for optical applications. For example, in some embodiments, the active layer **108** comprises indium gallium zinc oxide, indium tin oxide, indium gallium oxide, indium zinc oxide, indium gallium zinc tin oxide, indium oxide, or some other suitable material. In some embodiments, the active layer **108** is formed to have a thickness in a range of between, for example, approximately 3 angstroms and approximately 90 nanometers.

(45) As shown in cross-sectional view **800**A of FIG. **8**A, a passivation structure **114** is formed over

the active layer **108**. In some embodiments, the passivation structure **114** comprises a first passivation layer **116**. In some embodiments, the first passivation layer **116** is formed by way of a deposition process, such as chemical vapor deposition (CVD), for example. In some embodiments, the first passivation layer **116** may be formed in a chamber set to a temperature in a range of between, for example, approximately 200 degrees Celsius and approximately 400 degrees Celsius. In other embodiments, the first passivation layer **116** may be formed by some other deposition process such as, for example, PVD, ALD, or the like. In some embodiments, the passivation structure **114** has a thickness in a range of between, for example, approximately 5 nanometers and approximately 1000 nanometers.

(46) In some embodiments, the passivation structure **114** comprises a material that has a high film density, is hydrophobic, and is electrically insulating, such as oxygen-doped silicon carbide, nitrogen-doped silicon carbide, a mixture thereof, or the like. The passivation structure **114** prevents moisture (e.g., water) and oxygen from the environment from damaging the active layer **108**.

(47) FIG. **8**B illustrates a cross-sectional view **800**B of some other embodiments of forming the passivation structure **114** over the active layer **108**.

(48) As shown in the cross-sectional view **800**B of FIG. **8**B, in some other embodiments, the passivation structure **114** comprises a second passivation layer **202** arranged below the first passivation layer **116** such that the second passivation layer **202** is arranged between the first passivation layer **116** and the active layer **108**. Thus, in some embodiments, the second passivation layer **202** is formed over the active layer **108**, and then, the first passivation layer **116** is formed over the second passivation layer **202**. In some embodiments, the second passivation layer **202** is formed by way of a deposition process (e.g., PVD, CVD, ALD, etc.).

(49) In some embodiments, at least the first passivation layer **116** comprises a material that has a high film density, is hydrophobic, and is electrically insulating, such as oxygen-doped silicon carbide, nitrogen-doped silicon carbide, a mixture thereof, or the like. In some embodiments, the second passivation layer **202** may comprise a material that has a high film density, is hydrophobic, and is electrically insulating, such as oxygen-doped silicon carbide, nitrogen-doped silicon carbide, a mixture thereof, or the like; or the second passivation layer **202** may comprise a material that has a lower film density and/or is less hydrophobic than the first passivation layer **116**, such as, for example, silicon dioxide, silicon nitride, aluminum oxide, hafnium oxide, hafnium zirconium oxide, or the like.

(50) FIGS. **9**A, **10**A, and **11**A illustrate some embodiments of the subsequent steps of a method of forming a bottom-gate TFT having a passivation structure **114** comprising the first passivation layer **116** after the formation of the passivation structure **114** in FIG. **8**A. Thus, in some embodiments, wherein the passivation structure **114** comprises the first passivation layer **116**, the method includes steps illustrated in FIGS. **5-7**, **8**A, **9**A, **10**A, and **11**A.

(51) FIGS. **9**B, **10**B, and **11**B illustrates some embodiments of the subsequent steps of a method of forming a bottom-gate TFT having a passivation structure **114** comprising the first passivation layer **116** and the second passivation layer **202** after the formation of the passivation structure **114** in FIG. **8**B. Thus, in some embodiments, wherein the passivation structure **114** comprises the first passivation layer **116** and the second passivation layer **202**, the method includes steps illustrated in FIGS. **5-7**, **8**B, **9**B, **10**B, and **11**B.

(52) Thus, as shown in cross-sectional view **900**A of FIG. **9**A and in cross-sectional view **900**B of FIG. **9**B, a masking structure **902** is formed over the passivation structure **114**. The masking structure **902** may be formed using photolithography and removal (e.g., etching) processes. In some embodiments, the masking structure **902** comprises a photoresist material or a hard mask material. In some embodiments, the masking structure **902** may comprise openings that expose upper surfaces of the first passivation layer **116**.

(53) As shown in cross-sectional view **1000**A of FIG. **10**A and in cross-sectional view **1000**B of

FIG. **10**B, a removal process is performed to form a first opening **1002** and a second opening **1004** in the passivation structure **114**. The first and second openings **1002**, **1004** of the passivation structure **114** are based on the openings in the masking structure **902**. In some embodiments, the first and second openings **1002**, **1004** extend completely through the passivation structure **114** to expose the active layer **108**. In some embodiments, the removal process of FIGS. **10**A and **10**B comprise a dry etching process or a wet etching process. In some embodiments, the active layer **108** comprises a material that is substantially resistant to removal by the removal process of FIGS. **10**A and **10**B such that damage to the active layer **108** by the removal process of FIGS. **10**A and **10**B is mitigated. In some other embodiments (not shown), the removal process of FIGS. **10**A and **10**B may remove a small portion of the active layer **108** such that the first and second openings **1002**, **1004** extend below a topmost surface of the active layer **108**.

(54) Further, in some embodiments, the removal process of FIG. **10**B may comprise multiple etchants, wherein a first etchant is used to remove the first passivation layer **116**, and a second etchant is used to remove the second passivation layer **202**. In other embodiments, a same etchant may be used to remove the first and second passivation layers **116**, **202** in FIG. **10**B.

(55) As shown in cross-sectional view **1100**A of FIG. **11**A and in cross-sectional view **1100**B of FIG. **11**B, in some embodiments, a source contact **110** is formed in the first opening (**1002** of FIGS. **10**A and **10**B) of the passivation structure **114**, and a drain contact **112** is formed in the second opening (**1004** of FIGS. **10**A and **10**B). In some embodiments, the source contact **110** and the drain contact **112** are by depositing a conductive material over the active layer **108** and the passivation structure **114** followed by removing excess conductive material arranged over the passivation structure **114**. In some embodiments, the conductive material of the source and drain contacts **110**, **112** is deposited by way of a deposition process (e.g., PVD, CVD, PE-CVD, ALD, sputtering, etc.). In some embodiments, excess conductive material is removed by a removal process such as, for example, a planarization process (e.g., chemical mechanical planarization) or an etching process.

(56) In some embodiments, the source contact **110** is spaced apart from the drain contact **112** by the passivation structure **114**. Further, the source contact **110** and the drain contact **112** extend completely through the passivation structure **114** to contact the active layer **108**. In some embodiments, the source contact **110** and the drain contact **112** comprise a conductive material. In some embodiments, the conductive material may include a metal, such as, for example, copper, aluminum, tungsten, or the like. In some embodiments, the source contact **110** and the drain contact **112** comprise a conductive material that is also transparent such as, for example, indium tin oxide, doped zinc oxide, or the like.

(57) Thus, in some embodiments, the active layer **108** is covered by the passivation structure **114** and the source and drain contacts **110**, **112**. Because the passivation structure **114** comprises at least an upper portion that comprises a material that has a high film density, is hydrophobic, and is electrically insulating, the passivation structure **114** can mitigate moisture (e.g., water) and oxygen from entering the active layer **108**, thereby reducing damage to the active layer **108** and increasing the reliability of the bottom-gate TFT.

(58) FIG. **12** illustrates a flow diagram of some embodiments of a method **1200** of forming a bottom-gate TFT comprising a passivation layer that comprises silicon carbide.

(59) While method **1200** is illustrated and described below as a series of acts or events, it will be appreciated that the illustrated ordering of such acts or events are not to be interpreted in a limiting sense. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described herein. In addition, not all illustrated acts may be required to implement one or more aspects or embodiments of the description herein. Further, one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

(60) At act **1202**, a gate electrode is formed over a substrate. FIG. **5** illustrates a cross-sectional view **500** of some embodiments corresponding to act **1202**.

(61) At act **1204**, an active layer is formed over the gate electrode structure. FIG. **7** illustrates a cross-sectional view **700** of some embodiments corresponding to act **1204**.
(62) At act **1206**, a passivation structure is formed over the active layer. FIG. **8**A illustrates a cross-sectional view **800**A of some embodiments corresponding to act **1206**.
(63) At act **1208**, the passivation structure is patterned to form a first opening in the passivation structure and a second opening in the passivation structure. FIG. **10**A illustrates a cross-sectional view **1000**A of some embodiments corresponding to act **1208**.
(64) At act **1210**, a source contact is formed in the first opening, and a drain contact is formed in the second opening. FIG. **11**A illustrates a cross-sectional view **1100**A of some embodiments corresponding to act **1210**.
(65) FIGS. **13-19**B illustrate cross-sectional views **1300-1900**B of some embodiments of a method of forming a top-gate thin film transistor (TFT) having a passivation structure comprising silicon carbide. Although FIGS. **13-19**B are described in relation to a method, it will be appreciated that the structures disclosed in FIGS. **13-19**B are not limited to such a method, but instead may stand alone as structures independent of the method.
(66) As shown in cross-sectional view **1300** of FIG. **13**, in some embodiments, the active layer **108** is formed over the substrate **102**.
(67) As shown in cross-sectional view **1400** of FIG. **14**, in some embodiments, a continuous gate dielectric layer **1406** is formed over the active layer **108**, and a continuous gate electrode layer **1404** is formed over the continuous gate dielectric layer **1406**.
(68) As shown in cross-sectional view **1500** of FIG. **15**, in some embodiments, a removal process is performed to remove peripheral portions of the continuous gate dielectric layer (**1406** of FIG. **14**) and of the continuous gate electrode layer (**1404** of FIG. **14**) to form the gate dielectric layer **106** and the gate electrode **104**, respectively. Thus, in some embodiments, the gate electrode **104** is arranged over the active layer **108** and the gate dielectric layer **106**.
(69) In some embodiments, prior to the removal process of FIG. **15**, a masking structure may be formed over the continuous gate electrode layer (**1404** of FIG. **14**). In some such embodiments, the masking structure (not shown) may be formed using photolithography and removal (e.g., etching) processes. Then, in some embodiments, the removal process of FIG. **15** may remove portions of the continuous gate dielectric layer (**1406** of FIG. **14**) and the continuous gate electrode layer (**1404** of FIG. **14**) that do not directly underlie the masking structure. In some embodiments, the removal process of FIG. **15** may comprise a wet or dry etching process. In some embodiments, the removal process of FIG. **15** may comprise more than one etchant. Further, in some embodiments, the active layer **108** is substantially resistant to removal by the removal process of FIG. **15**.
(70) As shown in cross-sectional view **1600**A of FIG. **16**A, in some embodiments, a first passivation layer **116** is formed over the active layer **108** and the gate electrode **104**. In some embodiments, the first passivation layer **116** is at least as thick as a sum of the thicknesses of the gate electrode **104** and the gate dielectric layer **106**.
(71) In some embodiments, the first passivation layer **116** comprises a material that has a high film density, is hydrophobic, and is electrically insulating, such as oxygen-doped silicon carbide, nitrogen-doped silicon carbide, a mixture thereof, or the like. The first passivation layer **116** prevents moisture (e.g., water) and oxygen from the environment from damaging the active layer **108**.
(72) FIG. **16**B illustrates a cross-sectional view **1600**B of some other embodiments of forming the first passivation layer **116** over the active layer **108**.
(73) As shown in the cross-sectional view **1600**B of FIG. **16**B, in some other embodiments, a second passivation layer **202** is formed over the active layer **108** prior to the forming of the first passivation layer **116**. In some embodiments, the second passivation layer **202** is formed over the active layer **108** and the gate electrode **104**. Then, in some embodiments, a removal process (e.g., etching process) is performed to remove portions of the second passivation layer **202** that are

arranged over the gate electrode **104**. The remaining second passivation layer **202** still covers the active layer **108**. In some embodiments, the first passivation layer **116** is then formed over the second passivation layer **202** and the gate electrode **104**. In some embodiments, a sum of the thicknesses of the first and second passivation layers **116**, **202** are greater than or equal to a sum of the thicknesses of the gate electrode **104** and the gate dielectric layer **106**.

(74) FIGS. **17**A, **18**A, and **19**A illustrate some embodiments of the subsequent steps of a method of forming a top-gate TFT having a passivation structure **114** comprising the first passivation layer **116** after FIG. **16**A. Thus, in some embodiments, wherein the passivation structure **114** comprises the first passivation layer **116**, the method includes steps illustrated in FIGS. **13-15**, **16**A, **17**A, **18**A, and **19**A.

(75) FIGS. **17**B, **18**B, and **19**B illustrates some embodiments of the subsequent steps of a method of forming a top-gate TFT having a passivation structure **114** comprising the first passivation layer **116** and the second passivation layer **202** after FIG. **16**B. Thus, in some embodiments, wherein the passivation structure **114** comprises the first passivation layer **116** and the second passivation layer **202**, the method includes steps illustrated in FIGS. **13-15**, **16**B, **17**B, **18**B, and **19**B.

(76) As shown in cross-sectional view **1700**A of FIG. **17**A and in cross-sectional view **1700**B of FIG. **17**B, in some embodiments, a removal process is performed to remove portions of the first passivation layer **116** arranged over the gate electrode **104**. In some embodiments, the removal process of FIGS. **17**A and **17**B comprises an etching process or a planarization process (e.g., chemical mechanical planarization). In some embodiments, the gate electrode **104** and the first passivation layer **116** have substantially coplanar upper surfaces after the removal process of FIGS. **17**A and **17**B.

(77) After the removal process of FIG. **17**A, a passivation structure **114** comprising the first passivation layer **116** is arranged over the active layer **108** and surrounds the gate electrode **104** and gate dielectric layer **106**. After the removal process of FIG. **17**B, a passivation structure **114** comprising the first passivation layer **116** and the second passivation layer **202** is arranged over the active layer **108** and surrounds the gate electrode **104** and the gate dielectric layer **106**.

(78) As shown in cross-sectional view **1800**A of FIG. **18**A and in cross-sectional view **1800**B of FIG. **18**B, in some embodiments, a removal process is performed to form a first opening **1002** within the passivation structure **114** on a first side of the gate electrode **104** and a second opening **1004** within the passivation structure **114** on a second side of the gate electrode **104**. In some embodiments, the first and second openings **1002**, **1004** are formed by way of a masking structure (e.g., **902** of FIGS. **9**A and **9**B) and a removal process as described with respect to FIGS. **9**A, **9**B, **10**A, and **10**B. In some embodiments, the passivation structure **114** separates the gate electrode **104** from the first and second openings **1002**, **1004**.

(79) As shown in cross-sectional view **1900**A of FIG. **19**A and in cross-sectional view **1900**B of FIG. **19**B, in some embodiments, a source contact **110** is formed within the first opening (**1002** of FIGS. **18**A and **18**B), and a drain contact **112** is formed within the second opening (**1004** of FIGS. **18**A and **18**B). In some embodiments, the passivation structure **114** electrically separates the source contact **110** from the gate electrode **104** and the drain contact **112** from the gate electrode **104**.

(80) Thus, in some embodiments, the active layer **108** is covered by the passivation structure **114**, the source and drain contacts **110**, **112**, and the gate dielectric layer **106**. Because the passivation structure **114** comprises at least an upper portion that comprises a material that has a high film density, is hydrophobic, and is electrically insulating, the passivation structure **114** can mitigate moisture (e.g., water) and oxygen from entering the active layer **108**, thereby reducing damage to the active layer **108** and increasing the reliability of the top-gate TFT.

(81) FIG. **20** illustrates a flow diagram of some embodiments of a method **2000** of forming a top-gate TFT comprising a passivation layer that comprises silicon carbide.

(82) While method **2000** is illustrated and described below as a series of acts or events, it will be appreciated that the illustrated ordering of such acts or events are not to be interpreted in a limiting

sense. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described herein. In addition, not all illustrated acts may be required to implement one or more aspects or embodiments of the description herein. Further, one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

(83) At act **2002**, an active layer is formed over a substrate. FIG. **13** illustrates a cross-sectional view **1300** of some embodiments corresponding to act **2002**.

(84) At act **2004**, a gate electrode is formed over the active layer. FIG. **15** illustrates a cross-sectional view **1500** of some embodiments corresponding to act **2004**.

(85) At act **2006**, a passivation structure is formed over the active layer and laterally surrounding the gate electrode structure. FIG. **16**A illustrates a cross-sectional view **1600**A of some embodiments corresponding to act **2006**.

(86) At act **2008**, the passivation structure is patterned to form a first opening in the passivation structure on a first side of the gate electrode structure and a second opening in the passivation structure on a second side of the gate electrode structure. FIG. **18**A illustrates a cross-sectional view **1800**A of some embodiments corresponding to act **2008**.

(87) At act **2010**, a source contact is formed in the first opening, and a drain contact is formed in the second opening. FIG. **19**A illustrates a cross-sectional view **1900**A of some embodiments corresponding to act **2010**.

(88) Therefore, the present disclosure relates to a method of forming a thin film transistor comprising a passivation layer over an active layer, wherein the passivation layer has an upper portion that comprises a material having a high film density and is hydrophobic to prevent moisture and oxygen from the environment from diffusing through the passivation layer and damaging the active layer.

(89) Accordingly, in some embodiments, the present disclosure relates to a device comprising: an active layer arranged over a substrate; a gate electrode arranged over the substrate and spaced apart from the active layer by a gate dielectric layer; a passivation structure arranged over the active layer; a source contact extending through the passivation structure and contacting the active layer; and a drain contact extending through the passivation structure and contacting the active layer, wherein the passivation structure is hydrophobic.

(90) In other embodiments, the present disclosure relates to a device comprising: an active layer arranged over a substrate; a gate electrode arranged on a first side of the active layer and spaced apart from the active layer by a gate dielectric layer; a passivation structure arranged on the active layer; a source contact extending through the passivation structure to contact the passivation structure; and a drain contact extending through the passivation structure to contact the passivation structure, wherein an upper portion of the passivation structure comprises silicon carbide.

(91) In yet other embodiments, the present disclosure relates to a method comprising: forming an active layer over a substrate; forming a gate electrode over the substrate, wherein a gate dielectric layer separates the gate electrode form the active layer; forming a passivation structure over the active layer; patterning the passivation structure to form a first opening in the passivation structure and a second opening in the passivation structure, wherein the passivation structure comprises silicon carbide; and forming a source contact in the first opening and a drain contact in the second opening, wherein the source and drain contacts contact the active layer.

(92) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that

they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method comprising: forming an active layer over a substrate; forming a gate electrode over the substrate, wherein a gate dielectric layer separates the gate electrode form the active layer; forming a passivation structure over the active layer; patterning the passivation structure to form a first opening in the passivation structure and a second opening in the passivation structure; forming a source contact in the first opening and a drain contact in the second opening, wherein the source and drain contacts contact the active layer; and wherein the passivation structure comprises an upper portion and a lower portion arranged around the source and drain contacts, the upper portion including one or more of oxygen-doped silicon carbide and nitrogen-doped silicon carbide.
2. The method of claim 1, wherein a topmost surface of the gate electrode physically contacts a bottommost surface of the gate dielectric layer along an interface that continuously and laterally extends past opposing outermost sides of both the first opening and the second opening.
3. The method of claim 1, wherein the passivation structure has a maximum height that is less than or equal to a maximum height of the source and drain contacts.
4. The method of claim 3, wherein the passivation structure is formed to have topmost and bottommost surfaces laterally between the source contact and the gate electrode.
5. The method of claim 3, wherein the upper portion of the passivation structure comprises a mixture of both the oxygen-doped silicon carbide and nitrogen-doped silicon carbide.
6. The method of claim 3, wherein the passivation structure comprises a flat lower surface that continuously extends between a first sidewall of the passivation structure that forms the first opening and a second sidewall of the passivation structure that forms the second opening.
7. A method, comprising: forming one or more interconnects within a dielectric layer over a substrate; forming an active layer over the dielectric layer; forming a gate electrode over the dielectric layer, wherein a gate dielectric separates the gate electrode from the active layer; forming a passivation structure over the active layer; forming a source contact and a drain contact within the passivation structure, wherein the source contact and the drain contact are laterally separated by a part of the passivation structure; and wherein the passivation structure comprises an upper portion and a lower portion arranged around the source and drain contacts, the upper portion including one or more of oxygen-doped silicon carbide and nitrogen-doped silicon carbide.
8. The method of claim 7, wherein the lower portion of the passivation structure and the upper portion of the passivation structure laterally surround the source contact and the drain contact.
9. The method of claim 8, wherein the upper portion of the passivation structure extends from above a bottom of the source contact to a top of the source contact.
10. The method of claim 8, wherein the source contact and the drain contact vertically extend to a top of the passivation structure.
11. The method of claim 8, wherein a height of the source contact is substantially equal to a height of the passivation structure.
12. The method of claim 8, wherein the passivation structure is arranged along sidewalls of the gate electrode and the gate dielectric.
13. The method of claim 12, wherein the passivation structure laterally separates the gate electrode and the gate dielectric from the source contact and the drain contact.
14. The method of claim 12, wherein the passivation structure laterally and continuously extends from a sidewall of the gate electrode to a sidewall of the source contact.
15. A method, comprising: forming an active layer over a substrate; forming a gate electrode over the substrate, wherein the gate electrode is separated from the active layer by a gate insulator; forming a passivation structure over the active layer; forming a source contact and a drain contact

over the active layer, wherein the source contact and the drain contact extend from a top surface of the passivation structure to the active layer; and wherein the passivation structure comprises an upper portion and a lower portion arranged around the source and drain contacts, the upper portion including one or more of oxygen-doped silicon carbide and nitrogen-doped silicon carbide.

16. The method of claim 15, further comprising: forming an inter-level dielectric (ILD) layer over the substrate; forming an interconnect within the ILD layer; and wherein the gate electrode and the passivation structure are laterally separated from the interconnect by the ILD layer.

17. The method of claim 15, wherein the active layer comprises indium gallium zinc oxide, indium tin oxide, indium gallium oxide, indium zinc oxide, indium gallium zinc tin oxide, or indium oxide.

18. The method of claim 15, wherein the passivation structure physically contacts the active layer.

19. The method of claim 15, wherein the passivation structure physically contacts the active layer along an interface that is vertically below the gate electrode.

20. The method of claim 15, wherein the upper portion of the passivation structure contacts the lower portion of the passivation structure along an interface that continuously extends between the source contact and the drain contact.